

Chapter 1 – Matter on the Atomic Scale

1.1. What is Chemistry

1. Be able to identify, distinguish and define **macroscopic**, **microscopic**, and **atomic** scales for observation and measurement.

1.2. Classification of Matter

1. Be able to identify, distinguish, and define **element**, **atom**, **chemical compound**, **molecule**.
2. Be able to identify, distinguish, and define **pure substances**, **mixtures**, **homogeneous mixtures**, **heterogeneous mixtures**.
3. Be able to identify, distinguish, and define **physical properties**, **chemical properties**, **physical change**, **chemical change**.
4. Be able to identify, distinguish, and define the **physical states** of **solid**, **liquid**, **gas**.
5. Be able to identify and define **melting point** and **boiling point**.

1.3. Units and Measurements

1. Know the seven SI **base units** and their standard abbreviation for **length**, **mass**, **time**, **electric current**, **temperature**, **amount of substance**, **luminous intensity**.
2. Know the SI **derived unit** and its standard abbreviation for **volume**, **density**, **energy**.
3. Know name and standard abbreviation for the metric prefixes representing 10^{12} , 10^9 , 10^6 , 10^3 , 10^2 , 10^1 , 10^0 , 10^{-1} , 10^{-2} , 10^{-3} , 10^{-6} , 10^{-9} , 10^{-12} .
4. Understand **scientific notation**, be able to convert between decimal and scientific notation, and be able to perform calculations using numbers expressed in scientific notation.
5. Understand the meaning of **absolute zero**.
6. Be able to convert temperature measurements between the **Kelvin**, **Celsius**, and **Fahrenheit** temperature scales.
7. Be able to identify the number of **significant figures** in a number and in the result of an arithmetic calculation involving addition, subtraction, multiplication, and division.

1.4. Unit Conversion

1. Be able to use **conversion factors** in a **dimensional analysis** approach to convert units.
2. Know **density** is the ratio of mass to volume and be able to do calculations involving density.

2. Chapter 2 – Elements and Compounds

2.1. The structure of the atom

1. Know the charge, relative mass, symbol, and location within an atom of the three major subatomic particles, **proton**, **neutron**, and **electron**.

2. Know the components of an **atomic nucleus** and its location within the atom.
3. Be able to define an atomic mass unit and know its common abbreviations, **u** and **amu**.
4. Understand **atomic number** and **mass number** and how they are in an **atomic symbol**.
5. Be able to define **isotope** and understand what distinguishes isotopes.
6. Be able to use isotope **present abundance** (or **fractional abundance**) and exact isotope mass to determine atomic weight.
7. Be able to identify and distinguish **ions** and distinguish **cations** and **anions**.

2.2. Elements and the Periodic Table

1. Be able to recognize and identify **periods** and **groups** in the **periodic table**.
2. Using a periodic table, be able to identify the members and location within the periodic table of the named groups of **alkali metals**, **alkaline earth metals**, **pnictogens**, **chalcogens**, **halogens**, **noble gases**.
3. Using a periodic table, be able to identify the named regions of **main-group elements**, **transition metals**, **lanthanides**, **actinides**, **metals**, **nonmetals**, **metalloids**.
4. Be able to define and recognize **allotropes**.

2.3. Covalent Compounds

1. Be able to identify **covalent compounds** based on the bonding and distinguish between **molecular covalent** compounds and **network covalent** compounds.
2. Be able to define, use and distinguish **molecular formula**, **empirical formula**, **structural formula**, and condensed **structural formula**.
3. Be able to identify, interpret and distinguish **wedge-and dash**, **ball-and-stick**, and **space-filling** representations for molecules.
4. For **binary nonmetal covalent compounds**, be able to provide the **systematic name** given the formula and the formula given the systematic name.
5. For **inorganic acids**, be able to provide the systematic name when isolated or in aqueous solutions given the formula and the formula given the systematic name.
6. Be able to identify **oxoacids** and provide the systematic name given the formula and the formula given the name.

2.4. Ions and Ionic Compounds

1. Be able to **identify compounds** bonding based on the bonding.
2. Be able to predict the charge state of main-group atomic ions based on the group to which the element belongs and name the ion.
3. Understand the meaning of a Roman numeral when added the end of the element name in the name for a transition metal cation.
4. For the **polyatomic ions** listed in the Chapter 2 course notes, be able to provide the formula, including charge, given the name and the name given the formula.
5. For the **oxoanions** listed in the Chapter 2 course notes, be able to provide the formula, including the charge, given the name and the name given the formula.

6. Be able to use the charge on the cation and the anion of an ionic compound to predict the formula and provide the name. Given the name of an ionic compound, be able to provide the formula.
7. Be able to identify a compound as ionic or covalent given the name or the formula.